

Research



Cite this article: Garagash DI. 2021 Fracture mechanics of rate-and-state faults and fluid injection induced slip. *Phil. Trans. R. Soc. A* **379**: 20200129.

<https://doi.org/10.1098/rsta.2020.0129>

Accepted: 22 October 2020

One contribution of 13 to a theme issue
'Fracture dynamics of solid materials: from
particles to the globe'.

Subject Areas:

geophysics, mechanics, environmental
engineering

Keywords:

rate-and-state-dependent friction, fracture
mechanics, rupture fronts, equation of motion,
injection-induced slip

Author for correspondence:

Dmitry I. Garagash
e-mail: garagash@dal.ca

Electronic supplementary material is available
online at [https://doi.org/10.6084/m9.
figshare.c.5253505](https://doi.org/10.6084/m9.figshare.c.5253505).

Fracture mechanics of rate-and-state faults and fluid injection induced slip

Dmitry I. Garagash

Dalhousie University, Department of Civil and Resource
Engineering, Halifax, Canada

DIG, 0000-0002-0851-2655

Propagation of a slip transient on a fault with rate- and state-dependent friction resembles a fracture whose near tip region is characterized by large departure of the slip velocity and fault strength from the steady-state sliding. We develop a near tip solution to describe this unsteady dynamics, and obtain the fracture energy G_c , dissipated in overcoming strength-excursion away from steady state, as a function of the rupture velocity v_r . This opens a possibility to model slip transients on rate-and-state faults as singular cracks characterized by approximately steady-state frictional resistance in the fracture bulk, and by a stress singularity with the intensity defined in terms of $G_c(v_r)$ at the crack tip. In pursuing this route, we develop and use an analytical equation of motion to study 1-D slip driven by a combination of uniform background stress and a localized perturbation of the fault strength with the net Coulomb force ΔT . In the context of fluid injection, ΔT is a proxy for the injection volume V_{inj} . We then show that, for ongoing fluid injection, the propagation speed of a transient induced on a frictionally stable fault is bounded by a large-time limiting value proportional to the injection rate dV_{inj}/dt , while, for stopped injection, the maximum slip run-out distance is proportional to $V_{inj, total}^2$.

This article is part of the theme issue 'Fracture dynamics of solid materials: from particles to the globe'.

1. Introduction

Rate- and state-dependent friction law has been widely used to describe slip behaviour on the natural faults [1] and material interfaces [2]. Emerging from rock mechanics experiments [3–5], this modelling framework

embodies two fundamental mechanisms governing the resistance of an interface to sliding, namely, dependence of friction on the rate of sliding and on the continually accrued slip, the latter pacing evolution of the ‘state’ of the interface in response to changes in slip rate. Despite the framework’s limitations when applied to high (seismic) slip velocities,¹ the adoption of the rate-state friction in the numerical and analytical models over the last couple of decades has allowed the study of a wide spectrum of fault slip behaviour. This includes, on the one hand, studies of the earthquake nucleation process [9–11], seismic rupture styles and their transitions [12–14], and earthquake cycle [15,16], and, on the other, studies of slow slip transients which may arise as the result of a ‘failed’ earthquake instability on a frictionally unstable (rate-weakening) fault [17,18], or on presumably frictionally stable (rate-strengthening) parts of the fault in response to the loading from a near-by earthquake [19], or due to the fault strength reduction enacted by a fluid injection [20,21].

A wealth of numerical simulations of slip on rate-state faults, as well as laboratory observations of sliding on frictional interfaces ([22] and references therein), show that transient propagation of slip events bear resemblance to fracture, characterized by the leading edge well defined by locally very large but finite slip-rate gradient.² This fracture-like behaviour is not limited to seismic ruptures, but rather occurs across the spectrum of slip events from slow to fast. Examples of the fracture-like slow slip transients are shown in figure 1 for two scenarios of fluid injection in the vicinity of a frictionally-stable fault (see details in §2b). Slowly slipping patch/crack is seen to spread symmetrically along a fault from the hypocentral region, where pore pressure is perturbed by the fluid injection, and has decelerating/approximately steady long-term dynamics in the fixed-volume/fixed-volume-rate injection scenario.

Notwithstanding the diverse dynamics of slip, the transients have a well-defined leading edge (figure 1*a,a'*) characterized by large slip-rate gradient (c,c') and associated with the maximum value of the friction departure Δf from its instantaneous steady-state value (b,b'). Snapshots of the Δf -distribution along the fault allow us to identify (i) a relatively narrow, near tip ‘fracture process zone’ corresponding to a strong excursion of friction away from the steady-state ($\Delta f \gg b$), followed by (ii) an approximately steady-state condition ($\Delta f \approx 0$) in the bulk of the slipping patch, i.e. away from the fracture tip. It is this type of fracture-like behaviour that led Rice [23] to advocate for applying the ‘small scale yielding’ concept of fracture mechanics [24] to model rate-state slip transients as singular cracks characterized by the simplified, steady-state frictional resistance in the bulk of the fracture, and by a stress singularity at the fracture tip. The intensity of the crack tip singularity is defined by matching of the elastic energy release rate of the advancing crack, G , to the fracture energy G_c , defined here as the rate of the energy dissipation within the near tip process zone when overcoming the frictional resistance Δf in excess of the steady-state value. The resulting condition, $G(\ell, \text{loading}, \dots) = G_c(\dot{\ell}, \dots)$, can be regarded as the crack tip ‘equation of motion’ (EoM) [25], which can be solved for the tip evolution, $\ell = \ell(t)$, if the dependence of (i) G on the crack geometry (crack half-length ℓ), loading, and the steady-state frictional resistance in the bulk of the crack, and (ii) G_c on the characteristics of the near tip region, e.g. on the propagation speed $v_r = \dot{\ell}$, can be determined. Recently, [26] lent further support to the idea by explicitly identifying the energy release rate into the tip region of numerically simulated ruptures on a rate-state fault.

Systematic application of the outlined fracture mechanics concepts to the analysis of the rate-state frictional ruptures have been, however, hampered in part by the lack of the robust prediction of the fracture energy and its dependence on the rupture front speed v_r over the entire spectrum of slip, from slow to fast. Notwithstanding a suggestion [9,10,27] to use the frictional state evolution in response to a step-increase in slip velocity (approximating the passage of the rupture front) as a proxy to calculate the fracture energy. Application of this ad hoc approximation remains

¹When the shear-heating-driven thermal pressurization of pore fluid, microscopic flash heating of frictional asperities, and possibly macroscopic melting of the fault gouge [6–8] can dramatically alter the fault strength.

²In contrast to ‘pure’ fracture with its infinite displacement discontinuity gradient at the leading edge.

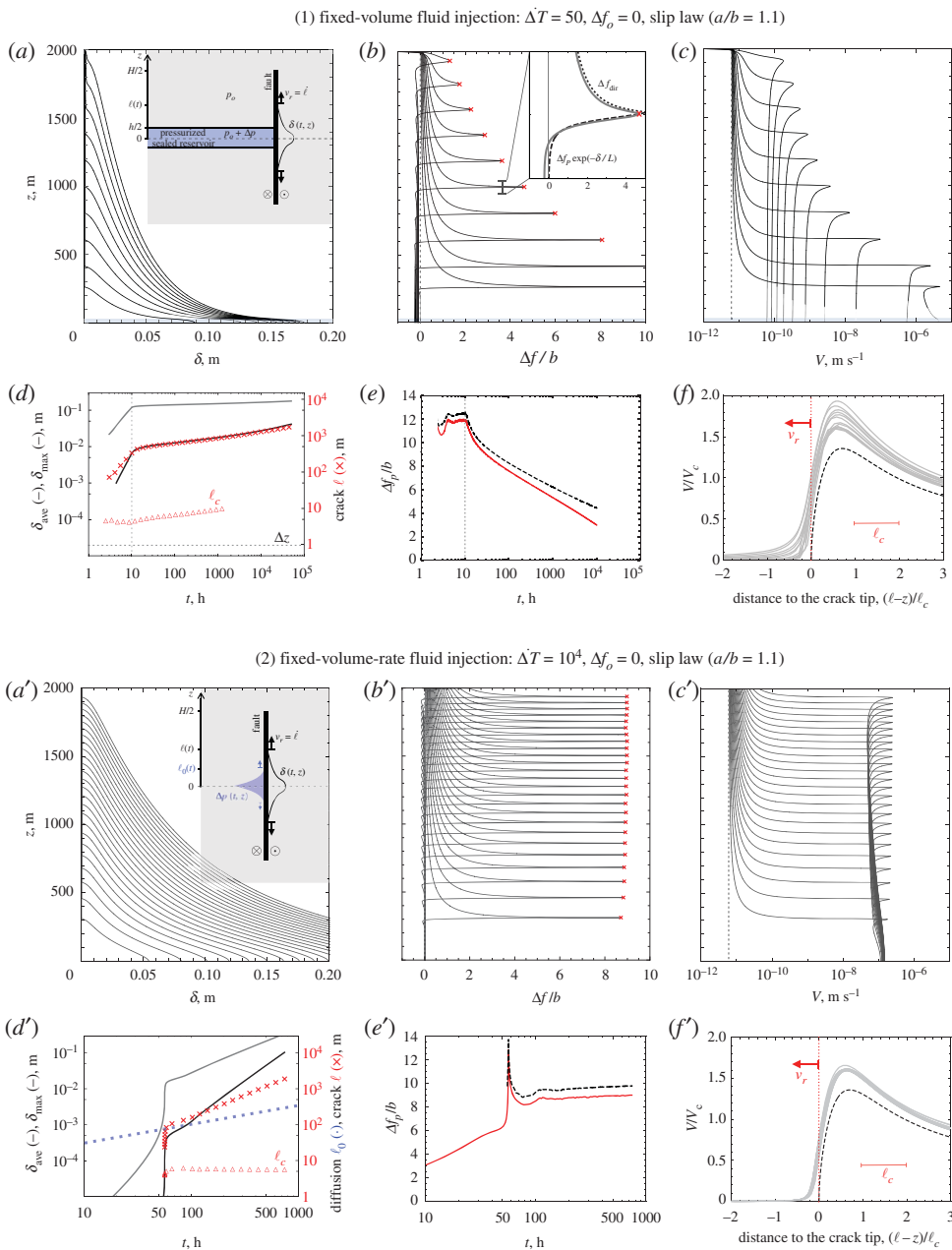


Figure 1. Symmetric slip development on a fault characterized by a rate-strengthening, rate-state friction in response to two scenarios: (i) a *fixed-volume* injection into a sealed-off reservoir abated by a fault ($a-f$) and (ii) a *fixed-volumetric-rate* injection into a permeable fault ($a'-f'$). Snapshots of slip, (a), departure of the stress ratio (friction) from the steady-state, $\Delta f = f - f_{ss}(V)$, (b), and slip rate, (c), are shown at 4 mm increments of the fault average slip $\delta_{ave}(t)$, (d). The front of the slipping patch (crack), $z = \pm \ell(t)$, is defined as the location of the peak friction change, Δf_p , shown by 'cross' in (b), while the evolution of the crack half-length ℓ (cross) and of the crack tip 'process zone' size ℓ_c (triangle) with time t shown in (d) and $\Delta f_p(t)$ in (e). Pore pressure diffusion front $\ell_0 = (4\alpha t)^{1/2}$ in the scenario 2, shown by dotted line in (d'), is outpaced by the slip front $\ell(t)$ after about 50 h of injection. Approximate collapse of the slip rate profiles from (c) when rescaled in the fracture tip scaling (V/V_c) and centred on the moving fracture tip (with the scaled distance to the tip $x = (\ell(t) - z)/\ell_c$), is shown in (f), where it is contrasted to the universal fracture tip solution (dashed line). (The scales V_c and ℓ_c are given by universal functions of the instantaneous rupture front speed $v_r = d\ell/dt$ as defined in the text). The fracture tip solution prediction of Δf_p is shown in (e) by dashed line. (Online version in colour.)

challenging³ given that the ‘slip velocity step’ is *a priori* unknown, and, furthermore, not well-defined, as the slip rate behind the rupture front may exhibit significant spatial and temporal variability (e.g. figure 1c,c’).

In this work, we overcome the limitations of the prior analyses by developing the universal solution for the near-tip region of the rate-state frictional rupture,⁴ which yields the dependence of the fracture energy G_c and of the peak stress perturbation carried by the rupture front on the instantaneous rupture front speed v_r . Importantly, rupture speed in the near tip solution can take on any value, from slow-aseismic to fast-seismic. We then complete the Equation of Motion, $G = G_c$, by developing the expression for the energy release rate G for a slip transient driven by a combination of two types of loading: (i) a uniform background overstress Δf_o (the initial fault stress-ratio in excess of the steady-state friction value at the ambient fault sliding velocity), and (ii) localized fault strength perturbation at the slip hypocentral region (due to, e.g. a fluid injection) quantified by the time-varying magnitude of the net Coulomb hypocentral force $\Delta T(t)$. General properties of slip transients, as predicted by the EoM and validated by a number of fully numerical slip simulations (such as in figure 1), are discussed when either the magnitude of the Coulomb force ΔT or its rate $\dot{\Delta T}$ is a fixed constant. Particularly, we show that the maximum run-out distance of the aseismic slip transient on a 1-D rate-strengthening fault is proportional to ΔT^2 when $\Delta T = \text{const}$, while the long-term run-out speed of the slip front is proportional to $\dot{\Delta T}$ when $\dot{\Delta T} = \text{const}$.

For fault slip driven by fluid injection, the assumption of the localized nature of strength perturbation stems from numerical modelling observations [20,21] and inferences from *in situ* experiments [29,30] that aseismic slip can significantly outpace pore pressure diffusion along the fault. The resulting Coulomb hypocentral force, $\Delta T \approx f_o \int \Delta p \, dz$ with $f_o =$ a representative value of friction there, is a simple proxy of (proportional to) the fluid volume injected into the fault zone—a result previously established by [31] for the particular case of a injection into a hydraulically sealed reservoir (such as Scenario 1 in figure 1, when no fluid diffusion along the fault is allowed), but shown here to be generally valid for arbitrary injection history into a permeable fault. As such, EoM allows us to track the maximum run-out distance of the aseismic slip in terms of the total injected fluid volume, and maximum slip front speed in terms of the average injection rate for an ongoing injection.

2. Fracture-like slip transients and behaviour near the fracture front

(a) Rate-state friction

The rate- and state-dependent friction can be written as [23]

$$f = f_{\text{dir}}(V) + \Theta, \quad f_{\text{dir}}(V) = f_o + a \ln \frac{V}{V_o}, \quad (2.1)$$

where $f_{\text{dir}}(V)$ is the so-called ‘direct-effect’ term, which governs the instantaneous response of friction to a change in the slip rate $V = \partial \delta / \partial t$ from the ambient fault state characterized by slip rate V_o and background shear-to-normal stress ratio f_o , while Θ is the so-called ‘state’ term, which governs the dependence of friction on history of slip. Evolution of the state Θ with either incurred slip δ or time t is driven by the departure of friction f from its instantaneous, slip-rate-dependent steady-state value $f_{\text{ss}}(V)$, denoted as⁵

$$\Delta f = f - f_{\text{ss}}(V), \quad (2.2)$$

³With an exception of the dynamic slip nucleation on a rate-weakening (ageing law) friction, when the EoM approach with the *ad hoc* approximation for the fracture energy correctly predicts the nucleation crack size [9,10].

⁴Of the kind used previously in [8,28] to determine fracture energy associated with co-seismic fault weakening by thermal pressurization and flash heating of frictional asperities.

⁵We note that similar quantities, $\Omega = \exp(\Delta f/b)$ [9,10] and $\Phi = 1 - \exp(-\Delta f/b)$ [11], have been previously used in the context of low-velocity (logarithmic) rate-state friction laws in order to track the friction’s departure from the steady state.

that can be generally written as

$$\frac{d\Theta}{d\delta} = -\frac{1}{L}\Psi(\Delta f). \quad (2.3)$$

Here, L is the state evolution distance (characteristic slip over which the change of the state takes hold) and function $\Psi(\Delta f)$ is such that $\Psi(0)=0$ and the sign of $\Psi(\Delta f)$ is that of Δf , or, in other words, a positive/negative change Δf of friction from the steady state leads to a decreasing/increasing Θ with slip, and, thus, a tendency of the friction to evolve towards the steady state.

State evolution function $\Psi(\Delta f)$ can be taken in either the ‘slip law’ or the ‘ageing law’ form⁶ [5]

$$\Psi(\Delta f) = \Delta f \quad (\text{slip law}), \quad \Psi(\Delta f) = b \left[1 - \exp\left(-\frac{\Delta f}{b}\right) \right] \quad (\text{aging law}). \quad (2.4)$$

The two laws are equivalent when that departure is small, $|\Delta f| \ll b$. The logarithmic dependence of the steady-state friction f_{ss} on the slip rate, $df_{ss}/dV = (a-b)/V$ has been suggested by *low velocity* laboratory frictional sliding experiments [3,4,32], and manifests either as a small rate-weakening (RW), when $a-b < 0$, or small rate-strengthening (RS), when $a-b > 0$, of the steady-state fault friction with $|a-b| \ll 1$. It will be convenient to write the steady-state friction expression in terms of the ambient fault state, V_o and f_o ,

$$f_{ss}(V) = (f_o - \Delta f_o) + (a-b) \ln \frac{V}{V_o} \quad (2.5)$$

where by definition (2.2), $f_o - \Delta f_o = f_{ss}(V_o)$ is the value of the steady-state friction at V_o , and Δf_o is the ambient value of the fault under (< 0) or over (> 0) stress ratio.

Rewriting (2.1) with the help of (2.2) as $\Delta f = \Delta f_{dir} + \Theta$, where $\Delta f_{dir} = \Delta f_o + b \ln(V/V_o)$, and then differentiating with regards to slip, we get

$$\frac{d\Delta f}{d\delta} = \frac{d\Delta f_{dir}}{d\delta} - \frac{\Psi(\Delta f)}{L}. \quad (2.6)$$

(b) Numerical modelling of injection-induced slip transients

Consider homogeneous ambient fault conditions, characterized by the background shear stress $\tau_{bg} = f_o \bar{\sigma}_o$, effective normal $\bar{\sigma}_o = \sigma - p_o$ stress (absolute normal stress minus pore pressure), slip rate V_o and initial value of the fault under/over stress ratio $\Delta f_o = f_o - f_{ss}(V_o)$. Slip transient propagation can take place from a hypocentral region centred at $z=0$, where the frictional strength $\tau = f \times \bar{\sigma}$ with $\bar{\sigma} = \sigma - p$ is perturbed by fluid injection. We neglect poroelastic stress perturbation [33], and focus on the direct effect of the pore pressure change $\Delta p(z, t) = p(z, t) - p_o$ on the strength. The spatial and temporal evolution of the pore pressure then fully describes a particular fluid injection scenario. To solve numerically for the induced slip transients, we couple the fault shear stress (strength) evolution, as governed by the rate-state friction law and by the pore pressure perturbation, with the elastic equation of stress transfer along a finite, $|z| < H/2$, fault in its quasi-dynamic form [16]. Further details of the numerical modeling are given in electronic supplementary material.

Numerical examples of the development of induced fault slip are shown in figure 1 for the following two scenarios of fluid injection into the vicinity of a fault characterized by a rate-strengthening, rate-state friction of the slip law type (electronic supplementary material, figure S1 shows the same scenarios in the case of the ageing law of rate-state fault friction).

Fixed-volume (stopped) injection, $V_{inj} = \text{const}$, into a hydraulically sealed reservoir layer abated by a fault (with no fluid leakage along the fault and out of the reservoir layer allowed), see

⁶Note that the ageing-law form of Ψ given here relies on the classical, low-velocity-type of the steady-state friction dependence on V , while the slip-law form of Ψ is applicable for any type of the steady-state friction dependence on V , including, e.g. high-velocity, flash heating type [7].

inset of figure 1a, results in a fixed pore pressure increase

$$\Delta p = \frac{V_{\text{inj}}}{\beta V_{\text{reservoir}}} \quad (2.7)$$

along the fault segment in contact with the reservoir, $|z| < h/2$, and zero outside of it (e.g. [31]), where β is the pore compressibility of the reservoir rock.

Continuing injection. $dV_{\text{inj}}/dt > 0$, directly into or nearby permeable fault damage zone,⁷ see inset of figure 1a', leads to the pore pressure diffusion along the fault with the diffusion front run-out distance $\ell_0 = \sqrt{4\alpha t}$ (e.g. [37,38]), where α is the hydraulic diffusivity of the fault rock. Pore pressure perturbation can be found by convolution

$$\Delta p(z, t) = \int_0^t \frac{dq}{dt'} p_G(z, t - t') dt' \quad (2.8)$$

of the injection rate⁸ $q(t) = (dV_{\text{inj}}/dt)/(w_{\text{FZ}}L_{\text{inj}})$ with Green's function, given by the $q = 1$ solution [39],

$$p_G(z, t) = \frac{1}{2\sqrt{\pi}} \frac{\ell_0(t)}{\alpha\beta} \Pi\left(\frac{|z|}{\ell_0(t)}\right) \quad \text{with } \Pi(\xi) = \exp(-\xi^2) - \sqrt{\pi}\xi \operatorname{erfc}(\xi).$$

For a *fixed-volume-rate injection*, $q = \text{const}$ and pore pressure perturbation is $\Delta p(z, t) = q \cdot p_G(z, t)$.

We use the following parametrization for examples in figure 1. Fault size $H = 4$ km, thickness of the sealed reservoir $h = 60$ m (scenario 1), fault discretization step $\Delta z = 2^{-13}H$; elasticity modulus $\mu = 30$ GPa; rate-state friction parameters: $a/b = 1.1$, $b = 0.01$, $L = 1$ mm, elasto-frictional length scale $\ell_b = (\mu/b\bar{\sigma}_o)L = 50$ m; initial fault state: $V_o = 6 \times 10^{-12} \text{ m s}^{-1}$, $\bar{\sigma}_o = \sigma_o - p_o = 60$ MPa, $f_o = 0.5 + \Delta f_o$, zero overstress $\Delta f_o = 0$. Perturbation in scenario 1: constant excess pore pressure $\Delta p = 50.7$ MPa (after 10 h ramp-up period) in the sealed reservoir part of the fault ($h = 60$ m). Perturbation in scenario 2: constant fluid injection rate $q = 3.7 \times 10^{-7} \text{ m}^3 \text{ m}^{-2} \text{ s}^{-1}$ per unit of the out-of-plane injection interval and unit thickness of the permeable fault zone characterized by the pore compressibility $\beta = 1/\text{GPa}$ and hydraulic diffusivity $\alpha = 0.01 \text{ m}^2 \text{ s}^{-1}$.

Generated slow slip transients spread crack-like, symmetrically along the fault (figure 1a,a'), and have decelerating/approximately steady long-term dynamics in the fixed-volume/fixed-volume-rate case (figure 1d,d'). In both cases, the slip front $z = \ell(t)$ outpaces the pore pressure perturbation front $z = \ell_0(t)$ (equal to $h/2$ in scenario 1 and $\sqrt{4\alpha t}$ in the scenario 2),

$$\ell(t) \gg \ell_0(t)$$

for all but early time period (approx. 10 h) of slip expansion out of the hypocentral region (figure 1d,d'). This localized nature of the pore pressure perturbation (compared to the size of the slipping patch) can be used to simplify the modelling by replacing the spatially-distributed shear strength perturbation by the net Coloumb force—the approach followed in the formulation of the approximate crack tip Equation of Motion in §3 and its application to the injection-induced slip in §4.

(c) Near rupture tip solution

Consider now evolution of slip near the 'front' of expanding slipping patch such as that exemplified by the numerical solutions shown in figure 1. Although such front is not mathematically sharp, it can be easily identified as a narrow band of very large but finite slip rate gradient (figure 1c) and a sharp peak in the friction departure Δf from the steady state (figure 1b). Zooming into the region near the rupture front, inset of figure 1b, shows that upon arrival of the front at a given point on the fault, the friction increases by the direct effect to the level significantly

⁷ ~ 1-to-100 metres thick zone of damaged/fractured rock and foliated gouges which embeds the ~ 0.01–1 metre thick fault's principal slipping zone (e.g. ultracataclastic gouge layer) [34]. The former contains fault rock units with relatively high permeability providing conduit(s) of the net thickness w_{FZ} for the flow of injected fluid along the fault, while the latter is characterized by very low permeability thus serving as a barrier to the fluid flow across the fault [35,36].

⁸ Per unit permeable fault zone thickness w_{FZ} and unit out-of-plane length of the fluid line-source, a segment of a horizontal injection wellbore parallel to the fault plane, L_{inj}

above the instantaneous steady-state value, until the fault has slipped a distance comparable to L and the emerging state-evolution reverses the friction towards the new steady-state value, i.e. towards $\Delta f = 0$. Denoting the corresponding peak change of friction from the steady state, Δf_p , the pre- and post-peak evolution of Δf near the rupture front are reasonably approximated by the direct-effect and state-evolution only, respectively,

$$\text{pre-peak: } \Delta f \approx \Delta f_{\text{dir}}(V) \quad \text{and} \quad \text{post-peak: } \frac{d\Delta f}{d\delta} \approx -\frac{\Psi(\Delta f)}{L} \quad (2.9)$$

(see the dotted and dashed lines in the inset of figure 1b, respectively).

The *post-peak* development of slip and the corresponding evolution of friction towards the steady-state attained away from the rupture front, but at distances still small compared to the rupture length, can be approximately modelled by a *semi-infinite crack* with the mathematically sharp tip ‘placed’ at the peak friction change Δf_p , *steadily* advancing with the rupture velocity v_r . (The latter given by the instantaneous value of the finite rupture solution). We will seek the steady tip solution for the slip $\delta(x)$, slip rate $V(x)$ and friction change $\Delta f(x)$ with the distance $x = v_r t - z$ from the moving crack tip. The elastodynamic equation of stress transfer for a semi-infinite, steadily propagating crack can be written in the coordinate x moving with the crack tip (see electronic supplementary material for details)

$$\Delta f(x) = \frac{\bar{\mu}(v_r)}{2\pi\bar{\sigma}} \int_0^\infty \frac{d\delta/dx'}{x' - x} dx', \quad V(x) = v_r \frac{d\delta}{dx}, \quad (2.10)$$

where $\bar{\mu} = \mu' g(v_r/c_s)$ is the apparent shear modulus which decreases with v_r from the quasi-static value μ' , given by μ for mode III and $\mu/(1-\nu)$ for mode II fracture, to zero when v_r is approaching a limiting velocity given by the shear wave speed c_s in mode III and the Raleigh speed c_R in mode II [25]. E.g. $g(v_r/c_s) = \sqrt{1 - (v_r/c_s)^2}$ in mode III.

Neglecting the pre-peak accumulation of slip (i.e. setting $\delta = 0$ and $\Delta f = \Delta f_p$ at the crack tip), integration of (2.9)₂ leads to the post-peak slip-weakening relation for Δf

$$\frac{\Delta f}{\Delta f_p} \approx \exp\left(-\frac{\delta}{L}\right) \quad (\text{slip law}), \quad \frac{\Delta f}{\Delta f_p} \approx \frac{b}{\Delta f_p} \ln \left[1 + \left(\exp \frac{\Delta f_p}{b} - 1 \right) \exp\left(-\frac{\delta}{L}\right) \right] \quad (\text{aging law}). \quad (2.11)$$

The ageing-law relation is parametrized by $\Delta f_p/b$. It can be shown, on one hand, to asymptotically coincide with the exponential slip-weakening form of the slip-law relation when $\Delta f_p/b \ll 1$, and, on the other, approach a linear slip-weakening $\Delta f = \Delta f_p - b\delta/L$ when $\Delta f_p/b \gg 1$ (figure 2a). Since slip-weakening relations (2.11) are independent of a particular time history of slip, they serve as a generalization of the similar relations obtained in [9] for a particular case of a step-increase of slip-rate in time.⁹

Both ageing- and slip-law weakening relations, (2.11), are of the exponential slip-weakening form for large enough slip $\delta \gg L$, but with different prefactors,¹⁰ which then leads to a well-defined fracture energy $G_c = \int_0^\infty \Delta f \bar{\sigma} d\delta$, as follows,

$$G_c = \Delta f_p \bar{\sigma} L \quad (\text{slip law}), \quad G_c = -\text{Li}_2 \left(1 - \exp \frac{\Delta f_p}{b} \right) \times b \bar{\sigma} L \quad (\text{aging law}), \quad (2.12)$$

where Li_2 is the polylog function [40]. It is instructive that for small $\Delta f_p/b$, the fracture energy of the ageing-law friction is asymptotically the same as that for the slip law and, therefore, linear in Δf_p , while for large $\Delta f_p/b$ the ageing-law fracture energy increases much more rapidly (quadratically) with Δf_p , $G_c/b\bar{\sigma}L = 0.5(\Delta f_p/b)^2$ (figure 2b).

The solution of the sharp-rupture tip problem (2.10)–(2.11) for the normalized friction change $\Delta f/\Delta f_p$, slip rate V/V_c , and slip δ/δ_c with the normalized distance from the tip x/ℓ_c is obtained using a method based on Gauss–Chebyshev numerical calculus [41] and shown in figure 3 for

⁹Eqns (6-7) of [9] are equivalent to (2.11) in this work in view of expression $\Delta f_p = b \ln(V_2/V_1)$ for the peak friction change from steady state for a velocity step from V_1 to V_2 in the former.

¹⁰For the ageing law, $\Delta f \approx b(\exp(\Delta f_p/b) - 1) \exp(-\delta/L)$ when $\delta/L \gg 1$.

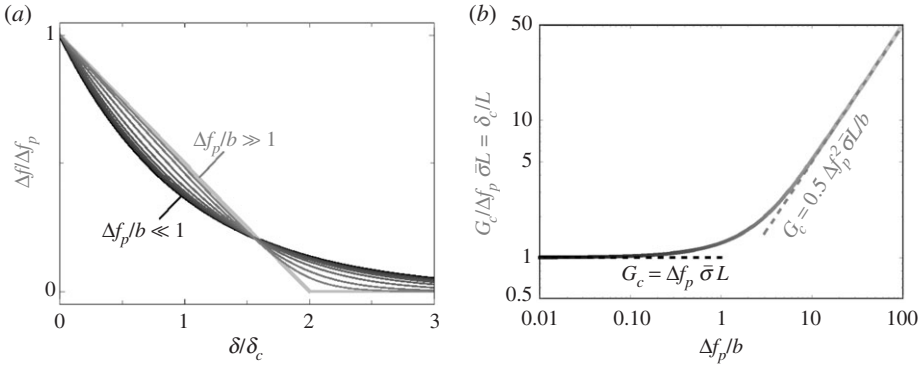


Figure 2. The post-peak slip-weakening near the rupture front, $\Delta f / \Delta f_p$ versus $\delta / \delta_c(\Delta f_p)$, for various $\Delta f_p / b$ values, (a), and normalized fracture energy $G_c / (\Delta f_p \bar{\sigma} L)$ (and slip weakening distance δ_c / L) as a function of $\Delta f_p / b$, (b), for the *ageing law* of rate-state friction. The post-peak evolution in the case of the *slip law* of rate-state friction for arbitrary $\Delta f_p / b$ is identical to the limit $\Delta f_p / b \ll 1$ of the ageing case, see solid and dashed black lines in (a) and (b), respectively.

the slip and ageing laws of the rate-state friction,¹¹ respectively. Here, the characteristic slip weakening distance δ_c , slip rate V_c and length scale ℓ_c are defined by

$$\delta_c = \frac{G_c}{\Delta f_p \bar{\sigma}}, \quad V_c = \frac{\Delta f_p \bar{\sigma}}{\bar{\mu}} v_r, \quad \ell_c = \frac{\bar{\mu}}{\Delta f_p \bar{\sigma}} \delta_c. \quad (2.13)$$

The normalized solution is parameter-less for the slip law, for which $G_c = \Delta f_p \bar{\sigma} L$ and $\delta_c = L$. In the case of the ageing law, the normalized solution is parameterized by $\Delta f_p / b$, since $\delta_c / L = G_c / (\Delta f_p \bar{\sigma} L)$ is an increasing function of $\Delta f_p / b$, see (2.12)₂, which is bounded by 1 and $0.5 \Delta f_p / b$ at small and large values of $\Delta f_p / b$, respectively. The slip rate (figure 3a) is seen to evolve with distance from the rupture front from the near, $V / V_c = k_0 (x / \ell_c)^{1/2}$ when $x \ll \ell_c$, to the far, $V / V_c = (2 / \pi^{1/2}) (\ell_c / x)^{1/2}$ when $x \gg \ell_c$, field asymptotes, while attaining the maximum value $\sim V_c$ at an intermediate distance $x \sim \ell_c$. The near-field prefactor k_0 in the ageing law solution is moderately varying with $\Delta f_p / b$ between 1.845 and 1.614 (inset in figure 3a), while it is a constant ($k_0 \approx 1.845$) in the slip-law case. The far-field corresponds to the classic singularity of the linear elastic fracture mechanics (LEFM) for a fracture characterized by a finite fracture energy G_c (e.g. [42]).

As evident from figure 3, length ℓ_c scales the size of the weakening region near the fracture front (the ‘process zone’). Expanding definition (2.13) of ℓ_c in terms of the fracture energy G_c and using the asymptotic behaviour of the latter, we can show that

$$\ell_c = \ell_b \cdot g(v_r / c_s) \cdot \begin{cases} (\Delta f_p / b)^{-1} & \left(\text{ageing law, } \frac{\Delta f_p}{b} \ll 1; \text{ slip law, arbitrary } \frac{\Delta f_p}{b} \right) \\ \frac{1}{2} & \left(\text{ageing law, } \frac{\Delta f_p}{b} \gg 1 \right) \end{cases} \quad (2.14)$$

is proportional to the elasto-frictional length scale $\ell_b = (\mu / b \bar{\sigma}) L$ (e.g. [10]). Where the first proportionality coefficient describes the dynamic contraction of the process zone (as g vanishes when v_r approaches the limiting dynamic speed) [42], while the second (in a curly bracket) signals very different size of the process zone for ruptures on faults with the ageing and slip laws of rate-state friction, respectively, particularly when $\Delta f_p \gg b$. Indeed, in this case, length scale ℓ_c for the slip law is a small fraction $2b / \Delta f_p$ of that for the ageing law. At 95% weakening (i.e. when Δf drops to 5% of its peak value), we have $x_{95\%} \approx 2.8 (\Delta f_p / b)^{-1} \ell_b$ and $0.7 \ell_b$ for the slip and ageing laws, respectively.¹² These results based on the universal near-rupture-tip solution are consistent

¹¹We note that the slip law solution is mathematically equivalent to the near-tip, underdrained-adiabatic solution for the rupture driven by the thermal pressurization obtained in [28].

¹²As obtained from figure 3b, $x_{95\%} \approx 2.8 \ell_c$ for slip and $1.4 \ell_c$ ageing laws, and equation (2.14)

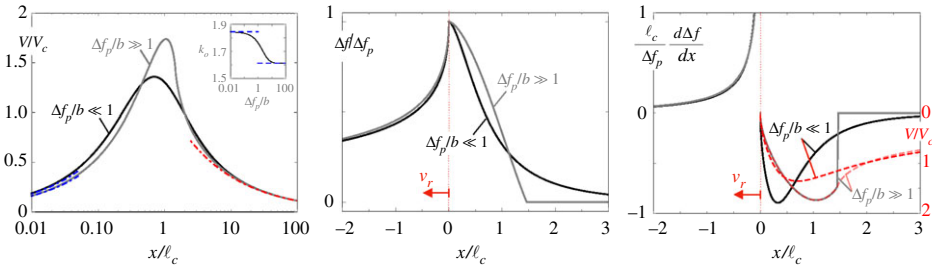


Figure 3. The post-stress-peak, sharp rupture tip solution for the normalized (a) slip rate V , (b) stress ratio change Δf , and (c) stress gradient $d\Delta f/dx$ with distance from the tip x for the *ageing law* of the rate-state friction in the limit of small ($\Delta f_p/b \ll 1$) and large ($\Delta f_p/b \gg 1$) peak friction departure from the steady-state. Dot-dashed lines in (a) show the near and the far fields of the slip rate, with the corresponding near field prefactor k_0 shown in the inset as a function of $\Delta f_p/b$. Dashed lines in (c) show the normalized slip rate from (a) to compare to the normalized stress gradient profiles. The rupture tip solution for the *slip law* of the rate-state friction for arbitrary value of Δf_p is given by the ageing law solution with $\Delta f_p/b \ll 1$. The length ℓ_c and slip rate V_c scales are defined in terms of the fracture energy (or, conversely, in terms of Δf_p) in (2.13). (Online version in colour.)

with the process zone size scaling inferred in [9] (and further discussed in [19]) from examples of fully numerical solutions of rate-state fault transients.

(d) Peak friction change and fracture energy

The post-peak, ‘sharp-tip’ solution allows us to determine the fracture energy G_c of the frictional process near the rupture front, (2.12), in terms of the peak value Δf_p of friction departure from the steady state. The goal of this section is to relate Δf_p , and by extension G_c to the rupture front velocity v_r .

The peak friction change Δf_p can be approximated by the pre-peak direct-effect friction evolution on one hand,

$$\Delta f_p \approx \Delta f_{\text{dir}}(V_p) = \Delta f_0 + b \ln \frac{V_p}{V_0}, \quad (2.15)$$

where V_p is the corresponding value of the slip rate, and by requiring $d\Delta f/d\delta = 0$ in (2.6) on the other,

$$b \left(\frac{d \ln V}{d\delta} \right)_{\text{at peak}} = \Psi \frac{(\Delta f_p)}{L}. \quad (2.16)$$

The solution of (2.15)–(2.16) for Δf_p and V_p depends on the pre-peak relation of the slip-rate to slip, and, in principle, requires the complete solution near the rupture front that couples the pre- and post-peak slip. To overcome the difficulty, we extrapolate the near-field behaviour obtained in the post-peak solution¹³ onto the immediate pre-peak region, thus allowing us to evaluate the left-hand side of (2.16) as $(k_0^2/2)(b/\delta_c)(V_p/V_c)^{-3}$. Accounting for the dependence (2.13) of V_c and δ_c scales on v_r and $G_c(\Delta f_p)$, equations (2.15)–(2.16) can be used to relate rupture velocity v_r and peak friction change Δf_p :

$$\frac{v_r/\bar{v}_0}{g(v_r/c_s)} = \mathcal{V}(\Delta f_p/b), \quad \text{with } \bar{v}_0 = \exp(-\Delta f_0/b) \frac{\mu'}{b\bar{\sigma}_0} V_0. \quad (2.17)$$

Here, $\bar{v}_0$ is a characteristic rupture velocity scale embodying the dependence on the fault ambient conditions, and function $\mathcal{V}(\Delta f_p/b)$ is given by

$$\mathcal{V}(\zeta) = \kappa_0 \frac{e^\zeta}{\zeta^{2/3}} \quad (\text{slip law}), \quad \mathcal{V}(\zeta) = \kappa_0(\zeta) \frac{-e^\zeta (1 - e^{-\zeta})^{1/3} \text{Li}_2^{1/3}(1 - e^\zeta)}{\zeta^{4/3}} \quad (\text{aging law})$$

¹³In the near-field ($x \ll \ell_c$) of the post-peak solution $V/V_c = k_0(x/\ell_c)^{1/2}$ and, by integration, $\delta/\delta_c = (2k_0/3)(x/\ell_c)^{3/2}$, thus, leading to slip to slip-rate relation $\delta/\delta_c = (2/3k_0^2)(V/V_c)^3$.

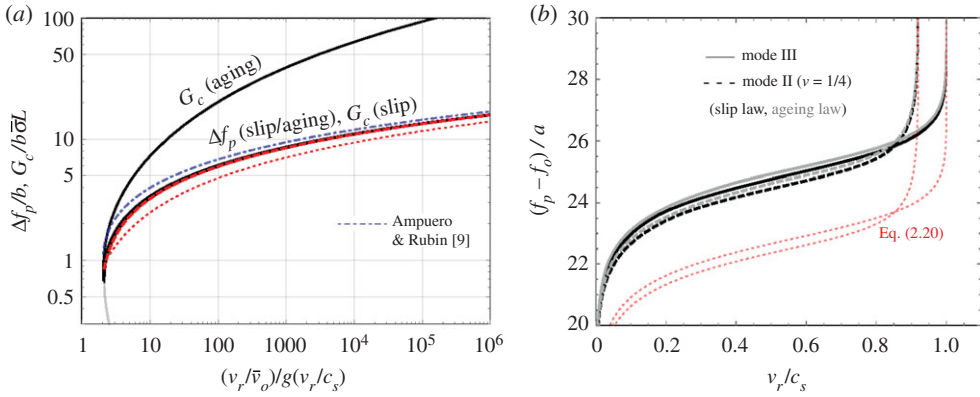


Figure 4. (a) Normalized peak friction departure from the steady state $\Delta f_p = (f - f_{ss}(V))_{\text{peak}}$ and fracture energy G_c as a function of the normalized rupture speed, $v_r/(\bar{v}_o g(v_r/c_s))$, where $\bar{v}_o = \exp(-\Delta f_o/b) (\mu'/b\bar{\sigma}) V_o$ embodies the dependence on the fault ambient conditions and $g(v_r/c_s)$ is the elastodynamic factor ($g(0) = 1$ and $g(1) = 0$ in mode III). The increasing/decreasing solution branch is shown by solid/opaque line. Red dotted and dashed lines show the simplified 0th (equation (2.19)) and 2nd iteration approximations of the increasing Δf_p solution, respectively. Blue dash-dotted line shows the relation based on eqns (45) and (53) of [9]. (b) Illustration of the resulting dependence of the 'at peak' stress-ratio referenced to the ambient value, $f_p - f_o$, on the rupture speed v_r/c_s in modes II (dashed) and III when $\bar{v}_o/c_s = 10^{-10}$ and $\Delta f_o = 0$. (Online version in colour.)

$\kappa_0 = (2/k_0^2)^{1/3}$ is a constant ≈ 0.838 for the slip law and is a function with values between $\kappa_0(0) \approx 0.838$ and $\kappa_0(\infty) \approx 0.916$ for the ageing law. The corresponding value of friction at the peak change follows from (2.2), $f_p = \Delta f_p + f_{ss}(V_p)$, and (2.15)

$$f_p = f_o + \left(\frac{a}{b}\right) (\Delta f_p - \Delta f_o) \quad (2.18)$$

Equation (2.17) provides an implicit form of the dependence of the peak friction change Δf_p , and, therefore, of the fracture energy G_c , (2.12), on the rupture speed v_r . This is plotted in figure 4a, showing $\Delta f_p/b$ and $G_c/b\bar{\sigma}L$ versus the scaled rupture velocity function¹⁴ $v_r/\bar{v}_o g(v_r/c_s)$. We note that these scaled relations are parameter-free (i.e. universal), and can be used to unravel the full dependence of the fracture energy on the problem parameters. The solution has distinct decreasing ($\Delta f_p/b \lesssim 1$) and increasing ($\Delta f_p/b \gtrsim 1$) branches, which characteristically correspond to distinct slip rate V_p values that are comparable to and larger than the fault ambient slip rate V_o , respectively. We anticipate that the underlining sharp-crack-tip model does not hold for the former decreasing branch, and henceforth consider the increasing Δf_p branch only. The solution is characterized by the minimum value of the fracture energy $G_{c,\min}/b\bar{\sigma}L = \Delta f_{p,\min}/b = 2/3$ attained at $v_{r,\min} \approx 2.138 \bar{v}_o$, and by the approximately-logarithmic increase of Δf_p for large values of $v_r/\bar{v}_o$. Interestingly, the Δf_p solutions for the slip and ageing laws are at most a few percent different, making them practically indistinguishable in log-scale of figure 4a. However, the fracture energy solutions for the two friction laws are drastically different when $v_r \gg \bar{v}_o$ (and $\Delta f_p \gg b$), with the ageing-law G_c -values much larger than the slip-law ones, owing to the different dependence (2.12) of the fracture energy on the peak friction change Δf_p for the two laws.

Figure 4b illustrates the corresponding dependence of the 'at peak' stress-ratio f_p , (2.18), on speed v_r of the mode II and III ruptures, respectively, for a particular fault ambient state. f_p increases with v_r and eventually diverges as v_r approaches the limiting value (c_s or c_R in modes III or II, respectively).

Equation (2.17) is implicit in $\zeta = \Delta f_p/b$, and convenient explicit approximations to its solution are obtained by applying an iterative process $\zeta^{(i+1)} = \ln(\mathcal{V}/\chi(\zeta^{(i)}))$ to invert $\mathcal{V}(\zeta) = \chi(\zeta)e^\zeta$ (where,

¹⁴For aseismic ruptures, $g \approx 1$ and the latter simply becomes a normalized rupture velocity $v_r/\bar{v}_o$.

e.g., $\chi(\zeta) = \kappa_0/\zeta^{2/3}$ for the slip law). Selecting the (seed) zeroth order approximation as $\zeta^{(0)}(\mathcal{V}) = \ln(\mathcal{V}/\kappa_0)$ yields an excellent approximation by the 2nd iteration, $\zeta^{(2)}(\mathcal{V})$ (figure 4a). Although $\zeta^{(0)}(\mathcal{V})$ provides a rather crude approximation (dotted red lines in figure 4), it does allow for a fair bit of interpretational simplicity. Indeed, in this approximation (2.17) reduces to

$$\Delta f_p \approx \Delta f_o + b \ln \frac{v_r/v_o}{\kappa_0 g(v_r/c_s)} \quad (\Delta f_p \gg b) \quad (2.19)$$

with the rupture velocity scale $v_o = (\mu/b\bar{\sigma})V_o$ is now determined by the fault ambient slip rate V_o only, and for the absolute value of the ‘at peak’ stress-ratio

$$f_p \approx f_o + a \ln \frac{v_r/v_o}{\kappa_0 g(v_r/c_s)} \quad (\Delta f_p \gg b). \quad (2.20)$$

Comparing (2.19) with (2.15), the quantity under the logarithm is the corresponding approximation of the slip rate at the peak friction change, i.e. $V_p/V_o \approx \kappa_0^{-1} v_r/(v_o g(v_r/c_s))$.

It is of interest to compare the universal solution for the fracture energy developed here to the corresponding expressions in [9,10]. For the slip-law friction, eqn (45) of Ampuero & Rubin [9] can be rephrased as $G_c = b\bar{\sigma}L \ln(V_{\max}/\bar{V}_o)$, where $V_{\max}$ is the maximum slip-rate and $\bar{V}_o = \exp(-\Delta f_o/b) V_o$. This relation translates to the one following from (2.12)₁ and (2.15) of this work if $V_{\max}$ is replaced by V_p . (The two slip rate values, although similar in magnitude, are quantitatively different with $V_p < V_{\max}$). These authors also propose a relation for the rupture propagation speed (their equation (53)), $v_r/\bar{v}_o = 0.75 (V_{\max}/\bar{V}_o) / \ln(V_{\max}/\bar{V}_o)$, based on a heuristic scaling of the numerical slip rate profiles in their study. The dependence of G_c on v_r following from the above two relations of [9] is shown by the blue dash-dotted blue line on figure 4a. It can be seen to provide an adequate approximation, particularly when $v_r \gg \bar{v}_o$, of the solution developed here.

The developed universal post-peak tip solution (figure 3) in the process zone scaling (2.13) and the corresponding solution for the peak friction change (figure 4a) are validated by comparing to the near tip region of full numerical solutions for the slow-slip transients on a fault with slip-law, rate-state friction, figure 1f,f' and 1e,e'. (Similar comparisons for other slow-slip numerical examples for the slip-law and ageing-law of fault friction are shown in electronic supplementary material, figures S1–S3). Particularly, the numerical slip rate profiles with peak velocities varying over several orders of magnitude over the event propagation time (figure 1c,c') are seen to collapse when using the tip scaling (2.13) and approximately match the universal tip solution profile (figure 1f,f'). The latter somewhat underestimates the former, but the agreement between the two improves with increasing rupture speed, or, conversely, deteriorates as the slip front and slip rate decelerate in the case of a constant hypocentral force (which also contributes to a moderate scatter of the rescaled numerical profiles, see also electronic supplementary material, figure S2 for examples of more ‘vigorous’ slow slip transients on an overstressed fault). Similarly, the peak friction change Δf_p , and hence the fracture energy, extracted from the numerical solutions compare well to, if being somewhat overestimated by, the rupture-speed-dependent prediction of the universal tip solution (figure 1e,e'). The approximate nature of the universal tip solution is likely a consequence of the assumptions taken (i.e. neglected direct effect post-friction-peak and the ‘sharp-crack-tip’ approximation, see further discussion in electronic supplementary material), which could be relaxed in future work.

3. Rupture front equation of motion

Examination of slip transients propagating on rate-state faults shows that the friction excursion away from the steady state is mostly confined to a relatively narrow region near the leading edge of the rupture, while the friction is approximately at the steady state within the rupture bulk. This suggests [23] that the rupture can be approximately modelled by that overcoming the steady-state frictional fault strength in the bulk of the slipping region and the finite fracture energy at

the advancing front, with the latter now an established function of the rupture velocity obtained from the universal near-tip solution.

(a) Fracture mechanics of slipping patch propagation

Propagation of a slipping patch within the above assumptions is governed by a variant of the small scale yielding fracture mechanics [24,25], stating that the energy release rate $G = \bar{K}^2/(2\bar{\mu})$ at the propagating crack tip matches the fracture energy $G_c = \Delta f_p(v_r)\bar{\sigma}_0 L$ obtained in the preceding section. The dynamic stress intensity factor $\bar{K}$ can be written as a product $\bar{K} = k(v_r)K$ of the static stress intensity factor K and a prefactor function $k(v_r)$ of the rupture velocity. The latter is only weakly dependent on the fracture configuration and the propagation history, and is approximated here by the form obtained from the self-similar, constant stress-drop symmetric crack expansion solution, which in the mode III case is given by $k(v_r) = \sqrt{1 - v_r^2/c_s^2}/E(1 - v_r^2/c_s^2)$, where E is the complete elliptic integral of the 2nd kind (e.g. [43]).¹⁵ Thus, we can write the propagation condition in terms of the static stress intensity factor as

$$K \approx K_c = \sqrt{\frac{2\mu'G_cg}{k^2}}, \quad (3.1)$$

where the fracture toughness K_c is a uniquely defined function of the rupture velocity $v_r = \dot{\ell}$.

We consider homogeneous ambient fault conditions, characterized by the background shear $\tau_{bg,0} = f_0\bar{\sigma}_0$ and effective normal $\bar{\sigma} = \bar{\sigma}_0$ stresses, over which the rupture (slipping patch) propagation takes place either quasi-statically (aseismically) or dynamically from a hypocenter at $z = 0$. In the hypocentral region, the fault loading and/or strength may be perturbed from the ambient values, which is presumed to give either initial or continuous stimulation of fault slip. E.g. it can be a temporal hypocentral loading leading to the nucleation of a run away dynamic rupture on a frictionally unstable fault (in which case it can be reasonably neglected for large time analysis of the rupture run-out), or it can be temporal or continual loading leading to aseismic slip propagation, as in e.g. localized reduction of fault strength by pore pressure increase due to fluid injection (as exemplified in §2b). In either case, we going to assume that the spatial extent of the perturbed hypocentral region is small compared to the slipping patch size, which lends itself to be modelled as a net Coulomb force $\Delta T(t)$ acting at the hypocenter. Due to the uniformity of the background loading and the point-force nature of the hypocentral perturbation, the problem is that of a symmetric bi-wing crack.

Consequently, we adopt the following approximate decomposition for the static stress intensity factor

$$K = K_{\Delta\tau} + K_{\Delta T}. \quad (3.2)$$

$K_{\Delta\tau}$ is the contribution to K due to the generally spatially variable stress drop $\Delta\tau = \tau_{bg} - f\bar{\sigma}$ along the bulk of the patch [24], away from the possibly perturbed hypocentral region, such that $\tau_{bg} = \tau_{bg,0} = f_0\bar{\sigma}_0$ and $\bar{\sigma} = \bar{\sigma}_0$, and away from the rupture front, such that $f \approx f_{ss}(V)$,

$$K_{\Delta\tau} = \sqrt{\frac{\ell}{\pi}} \int_{-\ell}^{+\ell} \frac{(f_0 - f_{ss}(V))\bar{\sigma}_0}{\sqrt{\ell^2 - z^2}} dz = \Delta\tau_{\text{eff}}\sqrt{\pi\ell}, \quad (3.3)$$

and the effective (equivalent uniform) stress drop $\Delta\tau_{\text{eff}}$ can be written in terms of the patch effective slip rate V_{eff} as

$$\Delta\tau_{\text{eff}} = (f_0 - f_{ss}(V_{\text{eff}}))\bar{\sigma}_0 \quad \text{with } f_{ss}(V_{\text{eff}}) \equiv \frac{1}{\pi} \int_{-\ell}^{\ell} \frac{f_{ss}(V)}{\sqrt{\ell^2 - z^2}} dz. \quad (3.4)$$

¹⁵To further substantiate the claim of the weak dependence of the $k(v_r)$ on a particular crack configuration, we observe that the expression $k(v_r) = \sqrt{1 - v_r^2/c_s^2}$ for a semi-infinite non-uniformly propagating mode III crack (e.g. [25]) is quantitatively only marginally different from the one adopted here.

$K_{\Delta T}$ is the contribution to K due to the hypocentral perturbation [44],

$$K_{\Delta T} \approx \frac{\Delta T}{\sqrt{\pi \ell}}, \quad (3.5)$$

where the net hypocentral Coulomb force

$$\Delta T = \int_{-\ell_0}^{+\ell_0} (\tilde{\tau}_{bg} - f_{ss}(V)\tilde{\sigma}) dz \approx \Delta T_{bg} - f_0 \Delta N \quad \text{with } f_0 = f_{ss}(V_{eff,0}) = \frac{1}{2\ell_0} \int_{-\ell_0}^{+\ell_0} f_{ss}(V) dz \quad (3.6)$$

embodies the effects of possible perturbations $\tilde{\tau}_{bg} = \tau_{bg} - f_0 \tilde{\sigma}_0$ and $\tilde{\sigma} = \tilde{\sigma} - \tilde{\sigma}_0$ ($= -\Delta p$) of the background shear and effective normal stresses, respectively, assumed to be mostly confined to a small hypocentral region $|x| < \ell_0$ with $\ell_0 \ll \ell$. An approximate expression for ΔT is given in (3.6) in terms of the assumed-to-be-known net background shear $\Delta T_{bg} = \int \tilde{\tau}_{bg} dz$ and net effective normal $\Delta N = \int \tilde{\sigma} dz = - \int \Delta p dz$ perturbative forces, and, in general, unknown value of the hypocentral friction f_0 .

Using (3.2) with (3.3) and (3.5), the propagation condition (3.1) can be written as

$$\Delta \tau_{eff} \sqrt{\pi \ell} + \frac{\Delta T}{\sqrt{\pi \ell}} = K_c \quad \text{with } \Delta \tau_{eff} = (f_0 - f_{ss}(V_{eff})) \tilde{\sigma}_0. \quad (3.7)$$

We next derive the bulk effective slip rate V_{eff} , defined in (3.4), as a function of ℓ and $\dot{\ell}$ by considering the slip rate distribution along the slipping patch, while similar derivation of the hypocentral slip rate $V_{eff,0}$ given in electronic supplementary material,¹⁶ and, thus, establish (3.7) as the rupture front equation-of-motion (EoM).

(b) Slip rate distribution and V_{eff}

In line with the crack loading decomposition, the slip along the patch is approximated by the superposition of that due to the effective uniform stress drop $\Delta \tau_{eff}$, $\delta_{\Delta \tau}$, and of that due to the hypocentral Coulomb force ΔT , $\delta_{\Delta T}$, (e.g. [44])

$$\delta \approx \delta_{\Delta \tau} + \delta_{\Delta T} \quad \text{with } \delta_{\Delta \tau} = \frac{2\Delta \tau_{eff}}{\hat{\mu}} \sqrt{\ell^2 - z^2}, \quad \delta_{\Delta T} = \frac{2\Delta T}{\pi \hat{\mu}} \cosh^{-1} \frac{\ell}{|z|}. \quad (3.8)$$

Here, $\hat{\mu}(v_r) = \mu' g(v_r)/k(v_r)$ is an apparent shear modulus which monotonically decreases with v_r from the quasi-static value μ' to a finite fraction thereof when v_r is approaching a limiting wave velocity [25,43].¹⁷ Specifically, for the mode III $g/k = E(1 - v_r^2/c_s^2)$, in which E is the complete elliptic integral of the 2nd kind, varies modestly between $(g/k)(0) = 1$ to $(g/k)(1) = 2/\pi$. The corresponding slip rate distribution follows by time differentiation¹⁸ of (3.8) and using (3.7)

$$V \approx \frac{2K_c \dot{\ell}}{\hat{\mu} \sqrt{\pi \ell}} \frac{1}{\sqrt{1 - z^2/\ell^2}} + \frac{2\dot{\Delta T}}{\pi \hat{\mu}} \cosh^{-1} \frac{\ell}{|z|}. \quad (3.9)$$

Consider first evaluating the effective slip rate V_{eff} , (3.4), when either the rate of change of the hypocentral Coulomb force or the fracture toughness is negligible. For the low-velocity

¹⁶Note that the dependence of $\Delta \tau_{eff}$ on V_{eff} is more essential to the accurate evaluation of EoM (3.7) than that of ΔT on $V_{eff,0}$, since the small changes of the former with effective slip rate are amplified in (3.7) by increasing function of crack length.

¹⁷This form of the apparent shear modulus is strictly valid for a self-similarly expanding crack with constant uniform stress drop. Here, we tentatively assume that the solution for the crack driven by a point-force can be formulated in terms of the same apparent modulus.

¹⁸Assuming a slowly varying stress drop $\Delta \tau$, toughness K_c , and apparent modulus $\hat{\mu}$, and, thus, neglecting their rates when carrying differentiation.

(logarithmically dependent on V) friction, we can get for the two end-members

$$V_{\text{eff}}^{(\dot{\Delta}T=0)} = \frac{4K_c \dot{\ell}}{\dot{\mu} \sqrt{\pi \ell}}, \quad V_{\text{eff}}^{(K_c=0)} = 0.4569 \times \frac{\dot{\Delta}T}{\dot{\mu}}. \quad (3.10)$$

In the general case, we can obtain from (3.4), after some manipulations,

$$V_{\text{eff}} = \mathcal{F}(\mathcal{P}) \frac{4K_c \dot{\ell}}{\dot{\mu} \sqrt{\pi \ell}}, \quad \text{where } \mathcal{P} = \frac{\dot{\Delta}T \sqrt{\pi \ell}}{K_c \dot{\ell}} \quad (3.11)$$

is the non-dimensional parameter with the meaning of the normalized point-force-rate. Function $\mathcal{F}(\mathcal{P})$, given by

$$\ln \mathcal{F}(\mathcal{P}) = \frac{1}{\pi} \int_{-1}^{+1} \ln \left(1 + \frac{\mathcal{P}}{\pi} \sqrt{1 - \xi^2} \cosh^{-1} \frac{1}{|\xi|} \right) \frac{d\xi}{\sqrt{1 - \xi^2}}, \quad (3.12)$$

and plotted in electronic supplementary material, figure S4, provides the transition with $\mathcal{P}$ from the toughness-dominated ($\mathcal{P} \ll 1$ or negligible $\dot{\Delta}T$), $\mathcal{F} \sim \exp(\mathcal{P}/\pi) \approx 1$, to the point-force-rate-dominated ($\mathcal{P} \gg 1$ or negligible K_c), $\mathcal{F} \sim 0.1142 \times \mathcal{P}$, regime. General expression (3.11) for the effective bulk slip rate naturally reduces to the asymptotic ones given in (3.10) when $\mathcal{P}$ is much smaller/larger than 1.

As a final note on the formulation of EoM, we acknowledge that the ambient slip rate on faults that are initially over- or understressed, i.e. when $\Delta f_0 \neq 0$, will evolve over a long (ambient) timescale $t_0 = L/V_0$ towards the new steady-state value, thus leading to reduction of the absolute value of Δf_0 over the same timescale. This baseline evolution of V_0 and Δf_0 can in principle be inserted into the EoM considerations without any change to the latter, but are practically unnecessary when considering transient evolution of ‘stimulated’ slip events on timescales shorter than t_0 (e.g. $t_0 \approx 5.3$ years for the fault parametrization in the numerical examples of slip transients considered in this work, such as in figure 1).

(c) EoM predictions

Let us make use of the following natural elasto-frictional scaling

$$\begin{aligned} \frac{t}{t_0} \rightarrow t, \quad \frac{\ell}{\ell_b} \rightarrow \ell, \quad \frac{v_r}{v_0} \rightarrow v_r, \quad \frac{V}{V_0} \rightarrow V, \quad \frac{\Delta T}{\mu' L} \rightarrow \Delta T, \\ \frac{\dot{\Delta}T}{\mu' V_0} \rightarrow \dot{\Delta}T, \quad \frac{G_c}{b \bar{\sigma}_0 L} \rightarrow G_c, \quad \frac{K_c}{b \bar{\sigma}_0 \sqrt{\ell_b}} \rightarrow K_c \end{aligned} \quad (3.13)$$

defined in terms of the characteristic length $\ell_b = (\mu'/b\bar{\sigma}_0)L$ and time $t_0 = \ell_b/v_0 = L/V_0$ scales. EoM (3.7), normalized with the help of (3.13) and divided through $\sqrt{\pi \ell}$, reads

$$\underbrace{\frac{\Delta f_0/b - (a/b - 1) \ln V_{\text{eff}}}{\Delta \tau_{\text{eff}}/(b \bar{\sigma}_0)}} + \frac{\dot{\Delta}T}{\pi \ell} = \frac{K_c}{\sqrt{\pi \ell}} \quad \text{with } V_{\text{eff}} = \frac{1}{g/k} \frac{4K_c \dot{\ell}}{\sqrt{\pi \ell}} \mathcal{F} \left(\frac{\dot{\Delta}T \sqrt{\pi \ell}}{K_c} \right), \quad (3.14)$$

where the scaled fracture energy $G_c = G_c(e^{\Delta f_0/b} \dot{\ell}/g)$ (figure 4) and toughness $K_c = \sqrt{2G_c g/k^2}$ are functions of the scaled rupture velocity $v_r = \dot{\ell}$ (while g and k are the previously defined elastodynamic functions of $v_r/(c_s/v_0)$ with $g = k = 1$ for aseismic slip). EoM is an implicit 1st order ordinary differential equation governing evolution of the rupture scaled half-length $\ell(t)$, which is parametrized by the direct-to-state friction parameters ratio a/b , ambient overstress $\Delta f_0/b$, scaled wave speed c_s/v_0 (only of importance for seismic ruptures), and scaled hypocentral force ΔT and its time-rate $\dot{\Delta}T$.

Consider the case of *constant hypocentral Coulomb force* ($\Delta T = \text{const}$, $\dot{\Delta}T = 0$, and $\mathcal{F}(0) = 1$) for which the EoM does not depend explicitly on time, and is entirely described by a trajectory in the phase-plane $(\ell, \dot{\ell})$. Such a trajectory, if need be, can be easily integrated to obtain rupture front progression with time $\ell = \ell(t)$. The EoM phase trajectories are shown in figure 5 for the

case of the slip-law friction¹⁹ and for systematically varied frictional rate-dependence parameter a/b (from rate-weakening, $a/b < 1$, to rate-strengthening, $a/b > 1$), hypocentral Coulomb force ΔT , and ambient fault overstress $\Delta f_o/b$.

For *frictionally stable faults* (neutral and RS panels of the figure), the overstressed conditions ($\Delta f_o/b = +1$) are seen to support long-term expansion and acceleration of slow slip with increasing run-out distance, such that the rupture speed v_r and the peak slip rate near the rupture front eventually approach seismic values, while the bulk slip rate V_{eff} approaches the steady state (aseismic) value corresponding to the ambient stress.²⁰ This eventual seismic transition on RS faults, appearing as a narrow seismic slip rate pulse, is predicted to occur over exceedingly large slip run-outs, which are likely not practical to observe in nature. In the case of a finite hypocentral perturbation ($\Delta T > 0$), early-time development of slip is dominated by a possibly seismic (for large enough ΔT) transient owing to an abrupt application of the Coulomb perturbation [19] followed by the rupture deceleration at intermediate times/lengths irrespective of the fault initial conditions (overstress value). This hypocentral-perturbation-dominated behaviour eventually gives way to the long-term slip accelerating dynamics on overstressed faults, as already described in the above, while the slip-rate fizzles to the ambient value (i.e. normalized bulk slip rate $V_{\text{eff}} \approx 1$) on under- and neutrally faults ($\Delta f_o/b \leq 0$), essentially signalling termination of the slip transient run-out.²¹ Taking $\Delta \tau_{\text{eff}} \approx \Delta f_o \bar{\sigma}_0$ and normalized $K_c \sim 1$ in the EoM leads to a simple quadratic equation for the maximum slip run-out distance ℓ_{max} . At zero fault overstress,

$$\Delta f_o = 0: \quad \ell_{\text{max}} = \Delta T^2 / 2\pi. \quad (3.15)$$

EoM examination of the slip development on *frictionally stable faults* with *ageing-law* of friction (phase plots on electronic supplementary material, figure S5) shows qualitatively similar but much slower dynamics owing to a much larger fracture energy when compared to the slip-law friction case at the same value of the rupture speed (figure 4).

For *frictionally unstable, RW faults* (left panel of the figure), a rather complicated variety of rupture behaviours is suggested by the EoM, including the non-unique rupture velocity (simultaneous existence of the accelerating and decelerating solution branches) for large enough rupture run-out distance. The critical run-out distance at which the non-uniqueness emerges may indicate the slip instability (earthquake nucleation) studied in the numerous previous works including [9–11]. Exploration of this possible connection is outside of the scope of this work, we do note however, that (i) care would need to be exercised in applying the crack-like EoM to earthquake nucleation on faults with slip-law rate-state friction, which were shown to support the pulse-like rupture propagation mode [9]; and, more optimistically, (ii) the critical slip run-out distance predicted by the EoM on initially steady-state faults ($\Delta f_o = 0$) with ageing-law of friction (top-left pane of electronic supplementary material, figure S5) is very well approximated by the earthquake nucleation length $\ell_{\infty} = \ell_b / (\pi(1 - a/b)^2)$ of [10,11].

In case of a *constant rate of the hypocentral Coulomb force* ($\Delta T > 0$), a useful approximation of the EoM (3.14) is obtained by differentiating in time and reasonably neglecting the rates of K_c and $\Delta \tau_{\text{eff}}$ (since both the toughness and the bulk stress drop are only weakly dependent upon the time-varying rupture velocity)

$$\underbrace{\frac{\Delta f_o}{b} - \left(\frac{a}{b} - 1\right) \ln V_{\text{eff}}}_{\Delta \tau_{\text{eff}} / (b \bar{\sigma}_0)} + \frac{\dot{\Delta T}}{\pi \ell} \approx \frac{1}{2} \frac{K_c}{\sqrt{\pi \ell}}. \quad (3.16)$$

When $\dot{\Delta T}$ is a constant, EoM (3.16) does not explicitly depend on time, and, thus, is fully described by a trajectory in the phase-plane. Figure 6 (electronic supplementary material, figure S6) shows

¹⁹Phase-plots for the ageing-law of friction are shown in electronic supplementary material, figure S5.

²⁰As also observed by [20] but in the context corresponding to a constant rate of Coulomb force (constant rate of fluid injection into the fault in that study).

²¹The slightly accelerating long-term rupture predicted by the EoM for the case of neutrally stressed fault $\Delta f_o = 0$ with $\Delta T > 0$ corresponds to the slip-rate in bulk of the rupture asymptotically approaching the ambient fault value, and, thus, an artefact of application of EoM outside of its underline assumptions.

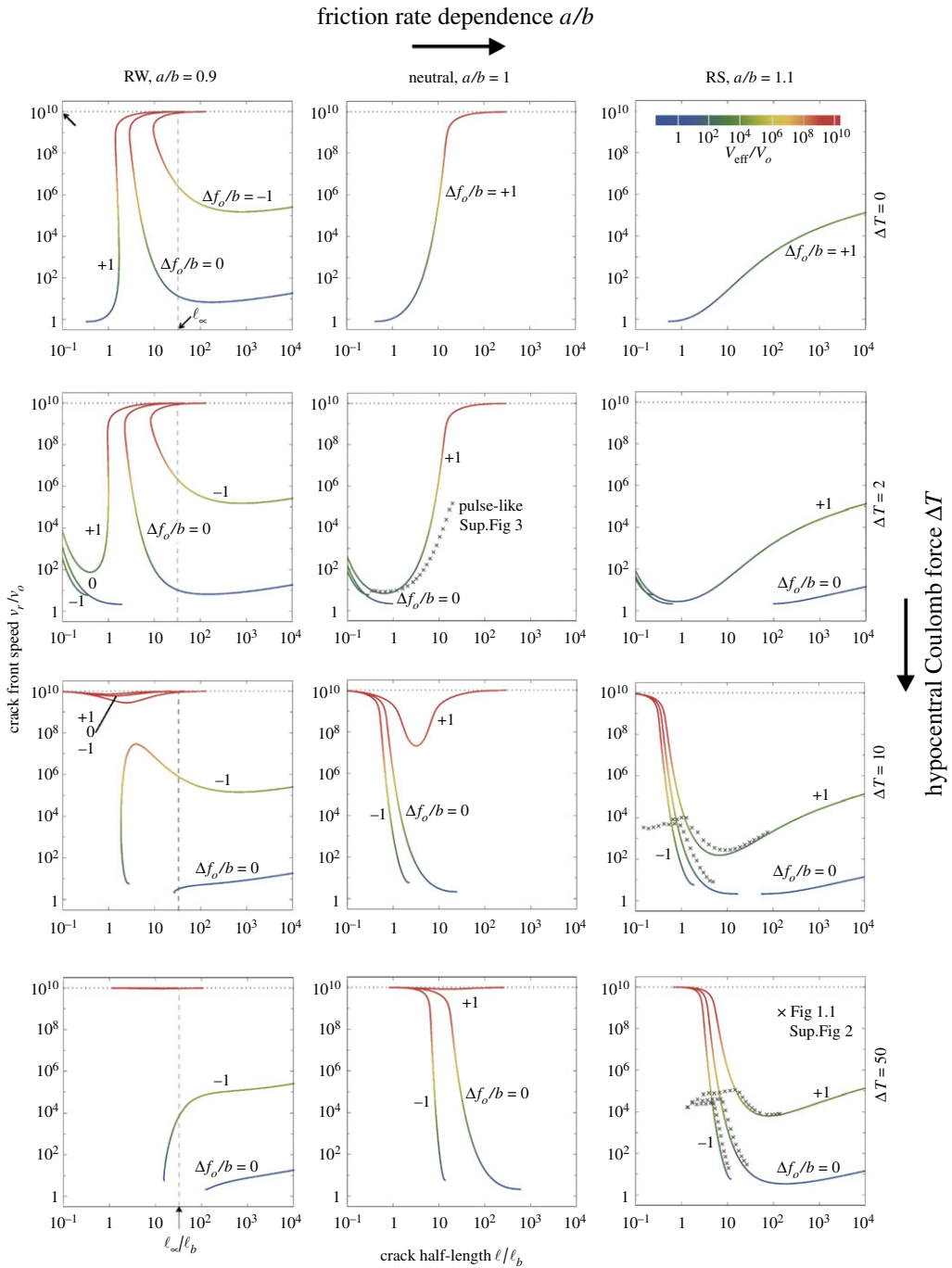


Figure 5. Equation of motion for *constant Coulomb hypocentral force*—phase plot of the rupture front speed v_r (scaled by $v_0 = (\mu/b\bar{\sigma})V_0$) versus half-rupture-length ℓ (scaled by $\ell_b = (\mu/b\bar{\sigma})L$) for a symmetric rupture on a fault with rate-state (*slip law*) friction characterized by rate-weakening (RW, left), rate-neutral (RN, middle) and rate-strengthening (RS, right) steady-state behaviour, and different initial overstress $\Delta f_o/b = -1, 0, +1$. Top to bottom panels correspond to different values of Coulomb force $\Delta T/\mu L = 0$ (top), 2, 10 and 50 (bottom). The effective slip rate magnitude within the bulk of the slipping patch is shown by colour gradient. Several examples of v_r versus ℓ evolution in full numerical solutions of fault slip transients driven by localized pore overpressure due to the fixed volume injection (such as in figure 1.1) are shown by ‘ $\times$ ’ for comparison. (Online version in colour.)

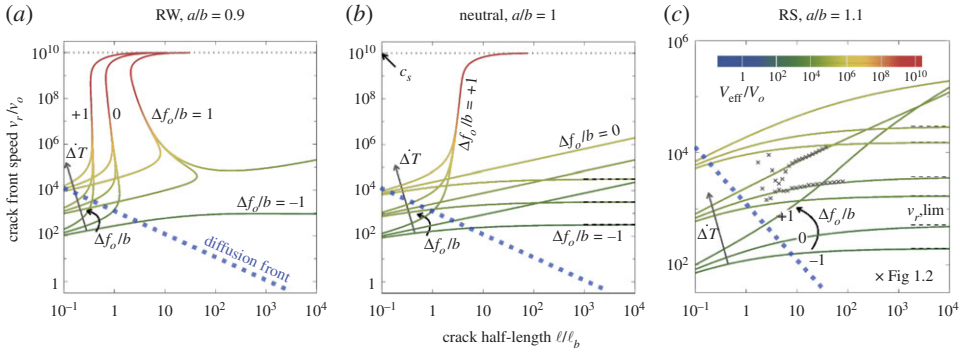


Figure 6. Equation of motion (3.16) for *constant rate of Coulomb hypocentral force*—phase plot of v_r versus ℓ for a symmetric rupture on a fault with rate-state (*slip law*) friction characterized by rate-weakening (a), rate-neutral (b) and rate-strengthening (c) behaviour, various initial overstress $\Delta f_o/b = -1, 0, +1$, and various values of the Coulomb force rate $\Delta T/\mu V_o = 10^3, 10^4, 10^5$. The effective slip rate magnitude within the bulk of the slipping patch is shown by colour gradient. Terminal (large run-out) rupture velocity $v_{r,\text{lim}}$ for the rate neutral (b) and strengthening (c) friction is shown by horizontal dashed lines. (Online version in colour.)

the EoM solution trajectories for the slip-law (ageing-law) friction, for a range of values of the rate-dependence parameter a/b , ambient overstress Δf_o and the rate of the hypocentral Coulomb force ΔT , respectively. Focusing on the response of *frictionally stable* faults (neutral and RS panels of figure 6) to the linear increase of the hypocentral Coulomb force with time, we observe that the initially overstressed faults ($\Delta f_o > 0$) support continually accelerating slip transients at large run-out distances similarly to the case with a constant Coulomb force (figure 5). This simply suggests that the fault overstress (when present), not the hypocentral perturbation, dominates the slip response at large time/run-out distances. However, the large-run-out rupture behaviour on neutrally- or under-stressed faults ($\Delta f_o \leq 0$) is markedly different from that for a constant Coulomb force, as the slip is seen to continuously accelerate towards the terminal rupture speed. This behaviour corresponds to the diminishing effect of the toughness compared to that of the Coulomb force-rate with increasing run-out distance, and bulk slip rate V_{eff} approaching the zero toughness limit given in (3.10). Correspondingly, zero-toughness reduction of the EoM can be solved for the normalized value of the terminal rupture speed

$$v_{r,\text{lim}} = \frac{\Delta T}{\pi} \left[\left(\frac{a}{b} - 1 \right) \ln V_{\text{eff}}^{(K_c=0)} - \frac{\Delta f_o}{b} \right]^{-1} \quad \text{with } V_{\text{eff}}^{(K_c=0)} = 0.4569 \Delta T \quad (3.17)$$

shown by dashed black lines on figure 6.

4. Implications for slip induced by fluid injection

The EoM can be applied to study the fluid-injection induced slip, e.g. in the context of the two scenarios discussed in §2b and figure 1, provided that the hypocentral Coulomb force (3.6) and its evolution can be evaluated based on the history of fluid injection and its associated pore pressure perturbation on the fault. In the absence of the shear stress perturbation, $\Delta T_{\text{bg}} = 0$, the Coulomb force is $\Delta T(t) = f_0 \int \Delta p(z, t) dz$ where the pore pressure perturbation is approximately confined to a hypocentral region along the fault, $|z| < \ell_0(t)$, with half-length $\ell_0(t)$.

In the scenario 1, when fixed fluid volume V_{inj} is injected into a hydraulically sealed reservoir ($V_{\text{reservoir}}$) abated by a fault, $2\ell_0 = h$ = the thickness of the reservoir and $\Delta T = f_0 \Delta p h$. In view of expression (2.7) for Δp ,

$$\Delta T = \frac{f_0 V_{\text{inj}}}{\beta V_{\text{reservoir}}/h} [\text{N m}^{-1}]. \quad (4.1)$$

In the scenario 2, when injecting directly into or nearby the permeable fault zone, $\ell_0(t) = \sqrt{4\alpha t}$ and the spatio-temporal evolution of the pore pressure perturbation is given by (2.8). Integrating the latter over the fault, we get a useful result that the Coulomb force is proportional to the currently injected fluid volume,

$$\Delta T(t) = \frac{f_0 V_{\text{inj}}(t)}{\beta w_{\text{FZ}} L_{\text{inj}}} [\text{N m}^{-1}], \quad (4.2)$$

irrespective of the injection history. Furthermore, this result is *not* restricted to the 1-D faults,²² such that, e.g. for fluid injection into a 2-D fault one finds $\Delta T(t) = f_0 V_{\text{inj}}(t)/(\beta w_{\text{FZ}}) [\text{N}]$.

The full numerical solution of slip transients in the two scenarios corresponding to fixed-volume (stopped) and fixed-volume-rate (continuing) injection are given in figure 1 for the case of zero ambient overstress $\Delta f_0 = 0$ and the choice of injection parameters for which the Coulomb force $\Delta T \approx 50 \times \mu L$ and the Coulomb-force-rate $\dot{\Delta T} \approx 10^4 \times \mu V_0$, respectively. When evaluating ΔT from (4.1) or (4.2) for these and other examples, we approximated the hypocentral friction $f_0 = f_{\text{ss}}(V_{\text{eff},0})$ by a representative constant independent of slip dynamics,²³ which particular value has been chosen as the mean $f_0 - \Delta f_0/2$ of the background stress-ratio value f_0 and the steady-state friction value at the ambient slip rate $f_{\text{ss}}(V_0) = f_0 - \Delta f_0$. (We note though that this approximation is likely to break down in the case of an earthquake slip instability, manifested by slip rate changes over many orders of magnitude, in which case rigorous evaluation of $f_0 = f_{\text{ss}}(V_{\text{eff},0})$ will be likely required, see electronic supplementary material). The numerical rupture dynamics in the two scenarios (see figure 1*d,d'* for the rupture half-length versus time) is contrasted to the EoM predictions in the phase plot of the rupture speed versus rupture half-length in figures 5 (bottom-right) and figure 6*c*, respectively, where the numerical solutions are shown by 'x'. We observe that EoM provides generally very good approximation of the decelerating (fixed-fluid-volume) and continuously accelerating (fixed-fluid-volume-rate) rupture dynamics in these two scenarios once the rupture is grown large enough compared to its hypocentral region. The EoM is seen to somewhat underpredict the rupture velocity in the fixed-volume case. This is attributed to a tendency of the direct friction effect to slightly reduce the friction in the bulk of a decelerating rupture from the steady-state value $f_{\text{ss}}(V_{\text{eff}})$ assumed in the EoM (electronic supplementary material, figure S8) and, thus, 'ease' the rupture propagation compared to the EoM prediction. We note that the EoM formulation can be extended to approximately account for the direct effect in the bulk (see details in electronic supplementary material), allowing for a more accurate prediction of rupture trajectories (electronic supplementary material, figure S9).

EoM also provides a good approximation of the rupture dynamics (figures 5 and 6) when injecting into initially overstressed ($\Delta f_0 > 0$), rate-strengthening faults (see electronic supplementary material, figure S2 for the full numerical solutions). The only attempted numerical example which the EoM failed to approximate corresponds to the accelerating aseismic rupture on overstressed ($\Delta f_0/b = 1$), rate-neutral fault ($a/b = 1$) with slip-law friction subject to perturbative force $\Delta T = 2$ (figures 5). In this case, slip propagates as a pulse rather than a crack (figure Figure 3*a*). The pulse-like propagation mode has been previously observed on rate-weakening, slip-law faults by [9]. When considering the same injection scenario and loading conditions, but on the ageing-law fault, the propagation is crack-like (electronic supplementary material, figure S3*a'*) and the EoM prediction is adequate (electronic supplementary material, figure S5).

Finally, we establish whether the EoM predictions are consistent with the underlining assumption of the localized nature of the pore pressure perturbation, $\ell_0(t) \ll \ell(t)$, or, conversely, forecast the conditions (e.g. on the fluid injection volume and/or the rate of injection) under which the slip propagation along the fault outpaces the pore pressure diffusion. The diffusion front $\ell_0 = \sqrt{4\alpha t}$ admits a simple 'trajectory' in the phase plane, $\dot{\ell}_0 \ell_0 = 2\alpha$, which is shown using scaling (3.13) by a dotted blue line on figure 6 for a particular value of the dimensionless diffusivity

²²and, in fact, follows simply from the fluid continuity and pore compressibility, yielding $\int_{\text{fault zone}} \Delta p \, dV = \beta V_{\text{inj}}$.

²³In lieu of the weak dependence of the steady-state friction f_{ss} on V , which typically amounts to a percent or less change in f_{ss} over few orders of magnitude change of V .

$\alpha/(\ell_b^2/t_0) = 0.66 \times 10^3$. (The latter value corresponds to, e.g. the parametrization of numerical examples of induced slip in §2b and figure 5, i.e. $\ell_b = 50$ m, $t_0 = 5.3$ yr, and fault zone diffusivity $\alpha = 0.01$ m² s⁻¹). Figure 6 clearly illustrates for the continuing injection case (linearly increasing Coulomb hypocentral force) that the slip front always eventually outpaces the decelerating diffusion front, while the time (run-out distance) it takes is decreasing with increasing injection rate $\propto \dot{\Delta T}$ and fault ambient overstress Δf_0 . For example, in the case $\dot{\Delta T}/\mu V_0 = 10^4$ of figure 6c and in the corresponding numerical example of the induced slip in figure 1.2, the slip runs out of the perturbed pore-pressure region at $\ell \sim \ell_b$.

5. Conclusion

In this work, we have developed a universal solution near the leading edge of a slip transient propagating on a fault with rate-state dependent friction. This ‘tip solution’ quantifies evolution of the slip velocity associated with the passage of the rupture front, characterized by the initial dramatic slip acceleration followed by deceleration towards the steady-state sliding in the bulk of the slipping patch. The corresponding maximum values of the slip velocity and shear stress perturbation carried by the rupture front (the latter related to the maximum excursion Δf_p of friction from the steady state) are established as functions of the instantaneous rupture front velocity v_r . The characteristic size $\ell_c(v_r)$ of this near tip process zone, shown to be a small fraction (e.g. given by $b/\Delta f_p(v_r)$ for the slip-law of the rate-state friction) of the classical frictional length scale $\ell_b = (\mu'/b\bar{\sigma})L$, provides an upper bound for the fault discretization size necessary to resolve slip transients by numerical approaches [16,19].

Developed tip solution allows for explicit evaluation of the fracture energy $G_c(v_r)$ dissipated within the near tip process zone when overcoming the frictional resistance Δf in excess of the steady-state value. This opens a possibility to model slip transients on rate-state faults as singular cracks characterized by the simplified, steady-state frictional resistance in the bulk of the fracture, and by a stress singularity at the fracture tip whose intensity is given in terms of $G_c(v_r)$. We have pursued this route and developed an analytical equation of motion for 1-D slip transients driven by a combination of a uniformly distributed background overstress Δf_0 and a localized fault strength perturbation at the hypocentral region, quantified by the net Coulomb force $\Delta T(t)$.

We have studied general properties of slip transients predicted by the equation of motion and further validated by a number of fully numerical slip simulations. Particularly, we have established that the maximum run-out distance of an aseismic slip transient on a rate-strengthening, 1-D fault is $\propto \Delta T^2$ when the Coulomb force ΔT is a constant ($\dot{\Delta T} = 0$), and that the long-term rupture speed $v_{r,\text{lim}}$ of the transients is $\propto \dot{\Delta T}$ when $\dot{\Delta T} > 0$. In the context of slip induced by fluid injection, the hypocentral Coulomb force is shown to be proportional to the injected fluid volume, e.g. $\Delta T(t) = f_0 V_{\text{inj}}(t)/\beta w_{\text{FZ}} L_{\text{inj}}$ in the case of injection into the permeable/damaged fault zone of thickness w_{FZ} over injection interval length L_{inj} , where β is the pore compressibility of the fault rock and f_0 a representative value of fault friction. The EoM solution then provides (i) a time-dependent upper bound for aseismic slip propagation on a frictionally stable fault $\ell_{\text{lim}}(t) \sim \int v_{r,\text{lim}} dt \propto V_{\text{inj}}(t)$ while the injection is ongoing, and (ii) the maximum run-out of the induced slip, $\ell_{\text{max}} \propto V_{\text{inj,total}}^2$, once injection has been stopped. The EoM predictions for realistic values of the fault and injection parameters support the idea of a localized pore pressure diffusion when compared with the aseismic slip run-out distance, and the related possibility of distant triggering of induced seismicity by the stress transfer driven by aseismic slip, rather than by the direct pore pressure increase by diffusion.

The EoM approach developed here under the uniform fault conditions is deemed translatable to heterogeneous faults with spatially variable background loading and frictional properties, provided that the length scale of the heterogeneity exceeds the size of the near tip process zone. The slip equation of motion and its possible extension to 2-D slip and to spatially heterogeneous faults provide a relatively simple analytical tool which can be applied to study a variety of transient fault slip phenomena, slow and fast, natural and induced by anthropogenic activities.

Data accessibility. This article does not contain any additional data.

Competing interests. I declare I have no competing interests.

Funding. The research has been supported by the Natural Science and Engineering Research Council of Canada, Discovery Program, grant no. 05743.

Acknowledgements. The author thanks two anonymous reviewers and Editor Koji Uenishi for their constructive comments.

References

1. Scholz CH. 2002 *The mechanics of earthquakes and faulting*, 2nd edn. Cambridge, UK: Cambridge University Press.
2. Baumberger T, Caroli C. 2006 Solid friction from stick–slip down to pinning and aging. *Adv. Phys.* **55**, 279–348. (doi:10.1080/00018730600732186)
3. Dieterich JH. 1979 Modeling of rock friction 1. Experimental results and constitutive equations. *J. Geophys. Res.* **84**, 2161–2168. (doi:10.1029/JB084iB05p02161)
4. Marone C. 1998 Laboratory-derived friction laws and their application to seismic faulting. *Annu. Rev. Earth Planet. Sci.* **26**, 643–696. (doi:10.1146/annurev.earth.26.1.643)
5. Ruina A. 1983 Slip instability and state variable friction laws. *J. Geophys. Res.* **88**, 10 359–10 370. (doi:10.1029/JB088iB12p10359)
6. Di Toro G, Han R, Hirose T, De Paola N, Nielsen S, Mizoguchi K, Ferri F, Cocco M, Shimamoto T. 2011 Fault lubrication during earthquakes. *Nature* **471**, 494–498. (doi:10.1038/nature09838)
7. Rice JR. 2006 Heating and weakening of faults during earthquake slip. *J. Geophys. Res.* **111**, B05311. (doi:10.1029/2005JB004006)
8. Viesca RC, Garagash DI. 2015 Ubiquitous weakening of faults due to thermal pressurization. *Nat. Geosci.* **8**, 875–879. (doi:10.1038/ngeo2554)
9. Ampuero J-P, Rubin AM. 2008 Earthquake nucleation on rate and state faults: aging and slip laws. *J. Geophys. Res.* **113**, B01302. (doi:10.1029/2007JB005082)
10. Rubin AM, Ampuero J-P. 2005 Earthquake nucleation on (aging) rate and state faults. *J. Geophys. Res.* **110**, B11312. (doi:10.1029/2005JA011064)
11. Viesca R. 2016 Self-similar slip instability on interfaces with rate- and state-dependent friction. *Proc. R. Soc. Lond. A* **472**, 20160254. (doi:10.1098/rspa.2016.0254)
12. Gabriel A-A, Ampuero J-P, Dalguer LA, Mai PM. 2012 The transition of dynamic rupture styles in elastic media under velocity-weakening friction. *J. Geophys. Res.* **117**, B09311. (doi:10.1029/2012JB009468)
13. Noda H, Dunham EM, Rice JR. 2009 Earthquake ruptures with thermal weakening and the operation of major faults at low overall stress levels. *J. Geophys. Res.* **114**, B07302. (doi:10.1029/2008JB006143)
14. Perrin G, Rice JR, Zheng G. 1995 Self-healing slip pulse on a frictional surface. *J. Mech. Phys. Solids* **43**, 1461–1495. (doi:10.1016/0022-5096(95)00036-I)
15. Lapusta N, Rice JR, Ben-Zion Y, Zheng G. 2000 Elastodynamic analysis for slow tectonic loading with spontaneous rupture episodes on faults with rate- and state-dependent friction. *J. Geophys. Res.* **105**, 23 765–23 789. (doi:10.1029/2000JB900250)
16. Rice JR. 1993 Spatio-temporal complexity of slip on a fault. *J. Geophys. Res.* **98**, 9885–9907. (doi:10.1029/93JB00191)
17. Liu Y, Rice JR. 2007 Spontaneous and triggered aseismic deformation transients in a subduction fault model. *J. Geophys. Res.* **112**, B09404. (doi:10.1029/2007JB004930)
18. Segall PA, Rubin AM, Bradley AM, Rice JR. 2010 Dilatant strengthening as a mechanism for slow slip events. *J. Geophys. Res.* **115**, B12305. (doi:10.1029/2010JB007449)
19. Perfettini H, Ampuero J-P. 2008 Dynamics of a velocity strengthening fault region: implications for slow earthquakes and postseismic slip. *J. Geophys. Res.* **113**, B09411. (doi:10.1029/2007JB005398)
20. Dublanchet P. 2019 Fluid driven shear cracks on a strengthening rate-and-state frictional fault. *J. Mech. Phys. Solids* **132**, 103672. (doi:10.1016/j.jmps.2019.07.015)
21. Eyre TS, Eaton DW, Garagash DI, Zecevic M, Venieri M, Weir R, Lawton DC. 2019 The role of aseismic slip in hydraulic fracturing-induced seismicity. *Sci. Adv.* **5**, eaav7172. (doi:10.1126/sciadv.aav7172)

22. Svetlizky I, Bayart E, Fineberg J. 2019 Brittle fracture theory describes the onset of frictional motion. *Annu. Rev. Condens. Matter Phys.* **10**, 253–273. (doi:10.1146/annurev-conmatphys-031218-013327)
23. Rice JR. 1983 Constitutive relations for fault slip and earthquake instabilities. *Pure Appl. Geophys.* **121**, 443–475. (doi:10.1007/BF02590151)
24. Rice JR. 1968 Mathematical analysis in the mechanics of fracture. In *Fracture, an Advanced Treatise* (ed. H Liebowitz), vol. II, ch. 3, pp. 191–311. New York, NY: Academic Press.
25. Freund LB. 1990 *Dynamic Fracture Mechanics*. Cambridge, UK: Cambridge University Press.
26. Barras F, Aldam M, Roch T, Brener EA, Bouchbinder E, Molinari J-F. 2020 The emergence of crack-like behavior of frictional rupture: Edge singularity and energy balance. *Earth Plan. Sci. Lett.* **531**, 115978. (doi:10.1016/j.epsl.2019.115978)
27. Hawthorne JC, Rubin AM. 2013 Laterally propagating slow slip events in a rate and state friction model with a velocity-weakening to velocity-strengthening transition. *J. Geophys. Res.* **118**, 3785–3808. (doi:10.1002/jgrb.50261)
28. Garagash DI. 2012 Seismic and aseismic slip pulses driven by thermal pressurization of pore fluid. *J. Geophys. Res.* **117**, B04314. (doi:10.1029/2011JB008889)
29. Bhattacharya P, Viesca RC. 2019 Fluid-induced aseismic fault slip outpaces pore-fluid migration. *Science* **364**, 464–468. (doi:10.1126/science.aaw7354)
30. Cappa F, Scuderi MM, Colletini C, Guglielmi Y, Avouac J-P. 2019 Stabilization of fault slip by fluid injection in the laboratory and in situ. *Sci. Adv.* **5**, eaau4065. (doi:10.1126/sciadv.aau4065)
31. Galis M, Ampuero J-P, Mai PM, Cappa F. 2017 Induced seismicity provides insight into why earthquake ruptures stop. *Sci. Adv.* **3**, eaap7528. (doi:10.1126/sciadv.aap7528)
32. Tullis TE. 1988 Rock friction constitutive behavior from laboratory experiments and its implications for an earthquake field monitoring program. *Pure Appl. Geophys.* **126**, 555–588. (doi:10.1007/BF00879010)
33. Segall P, Lu S. 2015 Injection-induced seismicity: poroelastic and earthquake nucleation effects. *J. Geophys. Res. Solid Earth* **120**, 5082–5103. (doi:10.1002/2015JB012060)
34. Chester FM, Evans JP, Biegel RL. 1993 Internal structure and weakening mechanisms of the San Andreas fault. *J. Geophys. Res.* **98**, 771–786. (doi:10.1029/92JB01866)
35. Faulkner DR, Jackson CAL, Lunn RJ, Schlische RW, Shipton ZK, Wibberley CAJ, Withjack MO. 2010 A review of recent developments concerning the structure, mechanics and fluid flow properties of fault zones. *J. Struct. Geol.* **32**, 1557–1575. (doi:10.1016/j.jsg.2010.06.009)
36. Wibberley CAJ, Shimamoto T. 2003 Internal structure and permeability of major strike-slip fault zones: the Median Tectonic Line in Mie Prefecture, Southwest Japan. *J. Struct. Geol.* **25**, 59–78. (doi:10.1016/S0191-8141(02)00014-7)
37. Ciardo F, Lecampion B. 2018 Effect of dilatancy on the transition from aseismic to seismic slip due to fluid injection in a fault. *J. Geophys. Res. Solid Earth* **123**, 1300–1315.
38. Garagash DI, Germanovich LN. 2012 Nucleation and arrest of dynamic slip on a pressurized fault. *J. Geophys. Res.* **117**, B10310. (doi:10.1029/2012JB009209)
39. Carslaw H, Jaeger JC. 1959 *Conduction of heat in solids*, 2nd edn. Oxford, UK: Oxford University Press.
40. Abramowitz M, Stegun IA (eds). 1965 *Handbook of mathematical functions*. New York, NY: Dover.
41. Viesca RC, Garagash DI. 2018 Numerical methods for coupled fracture problems. *J. Mech. Phys. Solids* **113**, 13–34. (doi:10.1016/j.jmps.2018.01.008)
42. Rice JR. 1980 The mechanics of earthquake rupture. In *Physics of the Earth's Interior, Proc. Int. Sch. Phys. Enrico Fermi*, 78 (eds AM Dziewonski, E Boschi), pp. 555–649. Elsevier.
43. Broberg KB. 1999 *Cracks and fracture*. Oxford, UK: Academic Press.
44. Tada H, Paris PC, Irwin GR. 2000 *The stress analysis of cracks handbook*, 3rd edn. New York, NY: ASME Press.